

Proposed Solution

AI-Assisted Smart Budget Planner

Project Proposal

PocketSmart AI transforms personal budget planning into a structured web workflow. It provides three specialized planners and combines deterministic budget packing with optional Gemini commentary.

Project Overview

Field	Details
Objective	Generate clear itemized budget plans for Home, Party and Jewelry scenarios while keeping the estimated total within the user's budget.
Scope	Authentication, three planners, optional Jewelry outfit image, saved history, merchant search links, optional Gemini insight and automated tests.
Problem Statement	Users need a simple way to allocate a fixed budget across multiple requested categories and preferences.
Impact	Provides a transparent estimated allocation, visible remaining amount and follow-up search links.
Approach	Use a shared catalog and deterministic whole-unit packing algorithm; use Gemini only for optional commentary.
Key Features	Registration/login; Home, Party and Jewelry planners; image validation; plan history; optional Gemini; responsive UI; automated tests.

Resource Requirements

Resource Type	Specification / Allocation
Software	Python 3.11+, FastAPI, Uvicorn, Pydantic v2, SQLite, PyJWT, HTTPX, Pytest.
AI	Optional Google Gemini API key.
Client	Modern web browser.
Deployment	Local by default; production requires HTTPS and secure cookie settings.